Computer Science

Summative Assignment

Module code and title	COMP4097 Advanced Computer Graphics and Visualisation
Academic year	2025-26
Coursework title	Visualisation
Coursework credits	5 credits
% of module's final mark	50%
Lecturer	George Koulieris
Submission date*	March 24th, 2pm
Estimated hours of work	10
Submission method	Ultra

Additional coursework files	Moon dataset, additional links including PyGMT tutorials and guidance.
Required submission items and formats	Please submit a compressed archive (.zip) with: (a) all your source code. Your program should run by simply calling `python3 problemX.py`, where `X` is 1 or 2 OR by opening a webpage if you embed the visualisation in an .html file.. Include a readme file with instructions on how to run your program and what external resources you require if this is not obvious. (b) A .pdf report no longer than 6 pages including images (max ~2500 words). At the top of the first page of the document identify yourself using your CIS username. (c) A short video demonstrating the two visualisations and interaction (compressed to <100MB).

* This is the deadline for all submissions except where an approved extension is in place.

Late submissions received within 5 working days of the deadline will be capped at 40%.
Late submissions received later than 5 days after the deadline will receive a mark of 0.
It is your responsibility to check that your submission has uploaded successfully and obtain a submission receipt.

Your work must be done by yourself (or your group, if there is an assigned groupwork component) and comply with the university rules about plagiarism and collusion. Students suspected of plagiarism, either of published or unpublished sources, including the work of other students, or of collusion will be dealt with according to University guidelines (https://www.dur.ac.uk/learningandteaching.handbook/6/2/4/).

COMP4097 Summative Assignment: Visualisation submodule

Lunar South Pole Visualisation for Artemis III

Important, please read first!

- The assignment should be submitted via Blackboard; the deadline is **March 24th, 2pm**.
- Your software will be tested and should work with Python 3.11.1. You can use additional Python libraries to achieve any additional functionality (e.g., NumPy). A requirements file must be provided with the tested version number.
- Please submit a compressed archive (.zip) with (a) all your source code. Your program should run by simply calling `python3 problemX.py`, where X is 1 or 2 OR by opening a webpage if you embed pre-rendered visualisations in an .html file. Include a readme file with instructions on how to run your program and what external resources you require if this is not obvious. (b) A .pdf report no longer than 6 pages including images (max 2500 words excluding references). At the top of the first page of the document identify yourself using your CIS username. (c) A short video demonstrating the two visualisations and interaction (compressed to <100MB).
- The marks available for correct implementations/answers to each question are indicated. Partial credit will be given for good attempts. Use of genAI is only allowed for research. The level of achievement (good/very good/excellent/etc.) for each marking criterion is determined based on the marking and classification conventions published in the university core regulations.
- The Visualisation sub-module contributes 50% of the total module mark and is only assessed by this coursework. (A FAQ section in Blackboard will be updated with questions *if* they arise.)

NASA's Artemis III mission aims to return humans to the surface of the Moon for the first time since Apollo, with a planned crewed landing at the lunar south pole in 2027. Unlike previous lunar missions, Artemis III will target a region characterised by extreme illumination conditions, rugged terrain, and permanently shadowed regions that may contain water ice. These factors introduce significant scientific, engineering, and operational challenges, but also offer unprecedented opportunities for long-term lunar exploration.

As a visualisation engineer supporting the Artemis III mission, you have been tasked with creating two interactive visualisations of the Moon's south polar region. The first visualisation is an exploratory tool for scientists and mission planners to gain insight into the lunar terrain and environmental conditions, aiding the identification of a safe and scientifically valuable landing site. The second visualisation will be used to communicate the goals, challenges, and importance of the Artemis III mission to the general public, helping to justify the cost and complexity of human space exploration to taxpayers.

The visualisations will be based on real lunar data collected by orbiting spacecraft, specifically surface elevation (heightmap), and illumination conditions & shadowed regions. You will combine these datasets to create compelling interactive maps of the lunar south pole. All data are provided in Blackboard. To visualise the data, you will be using Python and the open source library PyGMT. You can find a lot of helpful documentation in the PyGMT web pages, here[1] and here[2]. Your software or webpage should include a simple GUI to control certain aspects of the visualisation, as required by the problem descriptions below.

[1] https://www.pygmt.org/latest/
[2] https://forum.generic-mapping-tools.org/

PROBLEM 1, EXPLORATIVE VISUALISATION - 20 OUT OF 50 MARKS:

Create interactive maps of the Moon's south polar region (using the dataset provided) to effectively illustrate the terrain and environmental constraints relevant to selecting a landing site for the Artemis III mission. The visualisation should support scientific exploration of the region, helping mission planners identify locations that are safe for landing, while also providing access to solar illumination and proximity to shadowed regions that may contain water ice.

The user should be able to explore the data at multiple spatial scales, starting from a regional view of the lunar south pole and zooming into smaller areas by selecting regions of interest. The user should be able to sample relevant variables, such as elevation or illumination, by pointing to or clicking on the map. You will need to research and decide on the minimum & maximum elevation values corresponding to the pixel values of the height maps from the Lunar Reconnaissance Orbiter Camera (LROC). This functionality should enable detailed inspection of candidate landing areas and support informed decision-making (10 marks).

The visualised maps should compare different visualisation techniques for representing the lunar surface and associated variables. In particular, the visualisation should compare *3D perspective displacement map* rendering to using *isolines / isocontours*. The user should be able to modify parameters associated with each technique, for example for which heights isolines are generated. A graphical user interface should be provided to enable interactive map generation and parameter control (5 marks).

You may add any additional controls, overlays, or visual effects that you consider important to optimise the effectiveness of the visualisation in supporting scientific understanding of the lunar south pole. Part of the assessment is to demonstrate a creative and effective combination of different techniques to visualise the complex trade-offs involved in selecting a safe and suitable Artemis III landing site (5 marks).

PROBLEM 2, VISUALISATION FOR COMMUNICATION - 15 OUT OF 50 MARKS:

The second visualisation should be appropriate for a general audience. The interactive visualisation should combine elements of scientific visualisation, information visualisation, and infographics to convey key facts about the Artemis III mission and the lunar south pole environment. The visualisation should help a non-expert audience understand why the south pole was chosen as a landing site, what makes this region challenging to explore, and how factors such as sunlight and shadow influence mission design.

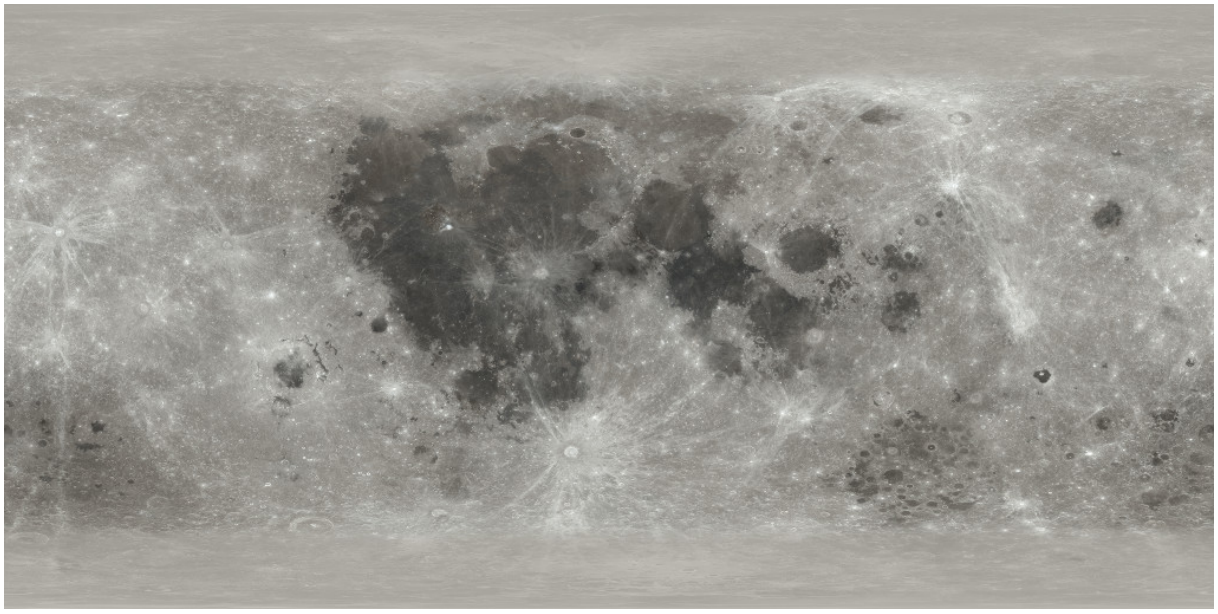
The choices made and approaches followed should be consistent with good practices for scientific and information visualisation, as discussed during the lectures. The visualisation should make effective and creative use of the provided lunar datasets while remaining accessible and easy to understand for the general public (10 marks).

A graphical user interface should be provided to enable interactivity, allowing users to explore different aspects of the visualisation as you see fit. As in the previous problem, part of the assessment is to demonstrate a creative combination of different visualisation techniques and narrative elements to effectively communicate the goals and significance of the Artemis III mission (5 marks).

PROBLEM 3, REPORT (& VIDEO) - 15 OUT OF 50 MARKS:

Compile a report no longer than six pages, including images, with a maximum of 2500 words excluding references, discussing the reasoning behind your choices for the above problems. At the top of the first page of the document, identify yourself using your CIS username.

For Problem 1, you should discuss the advantages and disadvantages of the different visualisation techniques used to represent the lunar terrain and environmental variables, i.e., three-dimensional terrain renderings versus contour-based representations. Include maps for different parts of the lunar south pole and at different spatial scales, and explain how these visualisations support scientific exploration and landing site selection for the Artemis III mission (8 marks).

For Problem 2, discuss your thought process behind the creation of your visualisation for the general public, explaining how you communicated the challenges and importance of lunar south pole exploration in an engaging and accessible way (7 marks).

Do not forget to attach a short video, compressed to less than 100MB, demonstrating your two visualisation techniques and their interactive features for a selection of parameters (**Students who do not submit a video will get a '0' mark for Problem 3**). Finally, select a candidate Artemis III landing site, show it in a screenshot, and briefly justify your choice.